

Chapter 1

Master Methodology Map

1.1 Introduction

This chapter describes the CTCC method at a conceptual level: what costs and benefits are included, how they depend on one another, and how they combine into a benefit–cost comparison. The aim is a single map of the *method*—no implementation details, only the ideas that define the analysis.

1.2 Visual Map of the Method

Figure 1.1 shows how the main concepts of the method fit together. Two kinds of preprocessing support the rest: **terrain-adjusted distance** (distance scaled by terrain difficulty) and **energy losses** (physical losses on the line). Terrain-adjusted distance feeds into construction and insurance costs; physical energy losses feed into line loss costs and emissions. Capital costs (construction, right-of-way, environmental mitigation) are followed by the revenue requirement (the rate-based transfer from ratepayers to the utility). Risk costs—insurance, delay, wildfire, and outage—come next. Benefits are congestion reduction, curtailment reduction, and delivered energy. Operations and maintenance costs, then emissions and line loss costs, complete the cost side. All costs and benefits are combined into a **benefit–cost ratio** (BCR), which can be computed under several perspectives (e.g. including or excluding certain risk or emissions components).

1.3 How the Method Fits Together

1.3.1 Preprocessing

The method uses two preparatory concepts. **Terrain-adjusted distance** is the line length scaled by terrain difficulty (e.g. mountain, river crossing). It is used to scale construction and insurance costs that are expressed per unit distance. **Energy losses (physical)** are the losses on the line and, for DC, in converters. They are computed from line and converter parameters and feed into the economic cost of losses and into emissions attributable to compensating generation.

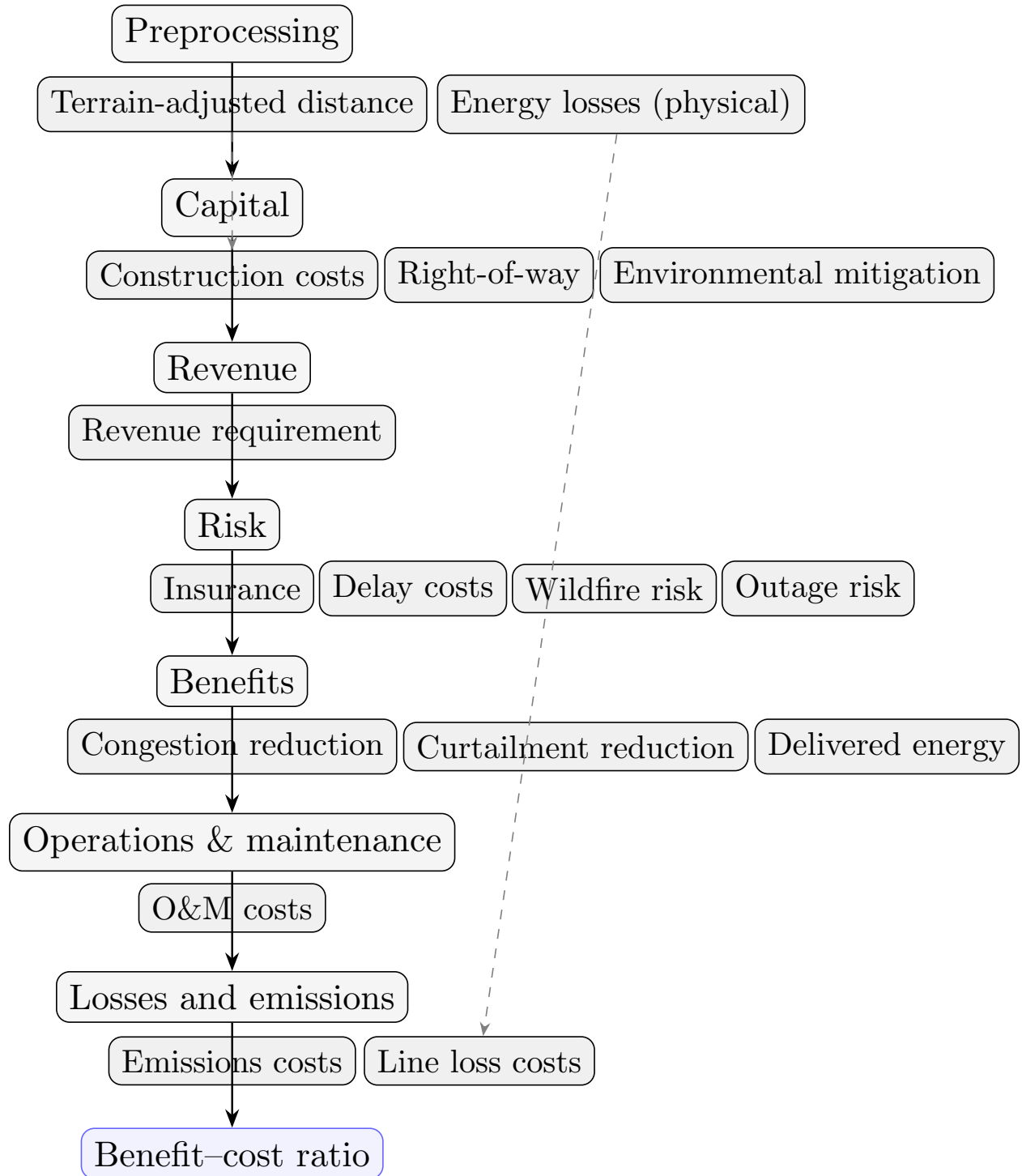


Figure 1.1: Conceptual flow of the CTCC method. Solid arrows show the main sequence; dashed arrows show key dependencies (terrain-adjusted distance into construction costs; physical energy losses into line loss costs).

1.3.2 Capital and revenue

Capital includes construction costs (conductors, structures, converters, adjusted for terrain and contingency), right-of-way costs, and environmental mitigation (temporary impacts and, for new build, wetland and habitat credits). These are combined with financing and timing to produce nominal, regulatory (AFUDC), and present-value measures. The **revenue requirement** is the rate-based revenue needed to recover capital and operating costs; it is a transfer between ratepayers and the utility, not a net social benefit, and is included only in stakeholder-specific benefit–cost views.

1.3.3 Risk

Risk costs include insurance (property, liability), construction and delay costs, wildfire risk (and liability where applicable), and outage risk. Each can be included or excluded depending on the perspective (e.g. societal vs. utility).

1.3.4 Benefits

Benefits are congestion reduction (value of reduced congestion from the project), curtailment reduction (value of reduced curtailment of generation), and the benefit of delivered energy (value of energy the line enables to be delivered each year, valued at an electricity price). Congestion and curtailment benefits can be scaled by conservative haircuts for societal BCRs. Revenue is not counted as a benefit in system-wide or societal perspectives.

1.3.5 Operations, losses, and emissions

Operations and maintenance costs cover vegetation management, conductor and structure O&M, and converter O&M. **Losses and emissions** include the economic cost of line losses (and converter losses for DC) and the cost of emissions from generation that compensates for those losses. Physical energy losses feed into both line loss costs and emissions.

1.3.6 Benefit–cost ratio

All cost categories (capital, operational, energy/emissions, risk, delay) and benefits (congestion reduction, curtailment reduction, and delivered energy) are combined into one or more **benefit–cost ratios**. Different BCR perspectives include or exclude specific cost or benefit components (e.g. wildfire, outage, emissions, line losses) so that analysts can compare full societal, risk-adjusted, or emissions-excluded views.

1.4 Concepts at a Glance

Table [1.1](#) summarizes each concept and its role in the method.

Concept	Role	Brief description
Terrain-adjusted distance	Preprocessing	Distance scaled by terrain difficulty; used to scale construction and insurance.
Energy losses (physical)	Preprocessing	Line and converter losses; feed line loss costs and emissions.
Construction costs	Cost	Capital cost of conductors, structures, converters (terrain- and contingency-adjusted).
Right-of-way	Cost	Cost of land and easements.
Environmental mitigation	Cost	Temporary impacts; wetland/habitat credits for new build.
Revenue requirement	Revenue	Rate-based revenue; transfer, not net benefit.
Insurance	Cost	Property and liability insurance.
Delay costs	Cost	Cost of construction delays.
Wildfire risk	Cost	Expected cost of wildfire risk (and liability if applicable).
Outage risk	Cost	Expected cost of outage risk.
Congestion reduction	Benefit	Value of reduced congestion.
Curtailment reduction	Benefit	Value of reduced curtailment.
Benefit of delivered energy	Benefit	Value of energy the line enables to be delivered (throughput).
O&M costs	Cost	Operations and maintenance (vegetation, conductors, structures, converters).
Emissions costs	Cost	Cost of emissions from loss-compensating generation.
Line loss costs	Cost	Economic cost of line (and converter) losses.
Benefit-cost ratio	Summary	Ratio of benefits to costs under chosen perspective.

Table 1.1: Concepts in the CTCC method and their role.